

Learning Intention: I can use the connection between addition and subtraction to find missing numbers.

Success Criteria — I can:

- Explain how addition and subtraction are inverse (opposite) operations.
- Use inverse operations to find an unknown number in a number sentence.
- Use inverse operations to check that my answer is correct.

Strategy Box

Addition and subtraction **undo each other** — they are inverse operations. If you know a part and the whole, you can find the missing part.

If $a + b = c$, then $c - b = a$ and $c - a = b$. Turn the number sentence around to find the missing number.

Worked Example

$$45 + \boxed{?} = 72$$

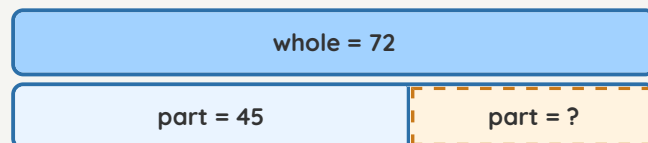
1. This is an addition sentence with a missing part. The **whole** is 72 and one **part** is 45.
2. To find the missing part, use the inverse operation: **subtract**.
3. $72 - 45 = 27$, so the missing number is **27**.
4. Check: $45 + 27 = 72$ ✓ It matches!

Review: Number Sentences

A **number sentence** uses numbers and symbols (+, −, =) to show a maths idea. An **unknown value** is a missing number, shown here as a box $\boxed{?}$. Your job is to work out what number belongs in the box.

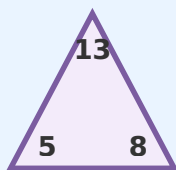
Part-Whole Model

The whole is made of two parts. If you know the whole and one part, subtract to find the other part.



Fact Families

A fact family shows how 3 numbers are connected through addition and subtraction. Knowing one fact helps you find the others.



$$\begin{aligned} 5 + 8 &= 13 \\ 8 + 5 &= 13 \\ 13 - 5 &= 8 \\ 13 - 8 &= 5 \end{aligned}$$

Parent Tip

Play "guess my number" at home: say a number sentence with a missing part out loud, like "6 plus what number makes 10?" Encourage your child to use subtraction to work it out, then check by adding.

Build Skills — Use inverse operations to find missing numbers.

Q1 — Missing Part (Addition Sentences)

Find the missing number in each sentence. Show your working.

Number sentence	Missing number	Check (inverse)
$14 + ? = 30$		
$? + 26 = 50$		
$37 + ? = 65$		

Q2 — Missing Part (Subtraction Sentences)

Find the missing number in each sentence. Show your working.

Number sentence	Missing number	Check (inverse)
$? - 18 = 24$		
$80 - ? = 35$		
$56 - ? = 19$		

Q3 — Complete the Fact Family

Use the three numbers to write all four number sentences in each fact family.

Numbers: 7, 9, 16

Numbers: 12, 8, 20

Q4 — Check the Working

Jayden solved these number sentences. Use the inverse operation to check his answers. Circle Correct or Incorrect.

Number sentence	Jayden's answer	Your check	Correct / Incorrect
$23 + 19 = 42$	42		Correct / Incorrect
$64 - 27 = 38$	38		Correct / Incorrect

Watch Out!

In a subtraction sentence like $? - 18 = 24$, the missing number is the **whole**, not a part — so you **add** to find it ($24 + 18 = 42$), even though the sentence uses a minus sign. Always ask: "Is the missing number the whole, or a part?"

Apply & Reasoning — Use what you know to explain your thinking.

- 1 Reasoning:** Sam had some marbles. He gave away 14 and has 23 left. Write a number sentence with a missing number, then solve it using inverse operations.

- 2 Reasoning:** Explain why subtraction can be used to check an addition answer. Use an example to help explain.

- 3 Open-ended:** Create your own missing-number sentence using two 2-digit numbers (like the examples on this worksheet). Show the missing number and how you found it.

Quick Check — Exit Ticket

Solve: $29 + ? = 41$. What is the missing number, and which operation did you use to find it?

Answer & Teacher Support

Answer Key — Pages 1 & 2

Worked example (p1): $45 + 27 = 72$; check $45 + 27 = 72$.

Q1 (p2): $14+16=30$; $24+26=50$; $37+28=65$.

Q2 (p2): $42-18=24$; $80-45=35$; $56-37=19$.

Q3 (p2): $7,9,16 \rightarrow 7+9=16$, $9+7=16$, $16-7=9$, $16-9=7$. $12,8,20 \rightarrow 12+8=20$, $8+12=20$, $20-12=8$, $20-8=12$.

Q4 (p2): $23+19=42$ — Correct (check: $42-19=23$). $64-27=38$ — Incorrect, correct answer is 37 (check: $64-27=37$).

Answer Key — Page 3

Q1: $? + 14 = 23$, so $? = 23-14 = 9$. Sam started with 9 marbles.

Q2: Subtraction "undoes" addition — if $a+b=c$, then $c-b=a$. E.g. $8+5=13$, and $13-5=8$ checks it.

Q3 (acceptable-answer guide): Accept any valid 2-digit + 2-digit missing-number sentence where the missing number is correctly calculated using the inverse operation and shown as working.

Exit ticket: $? = 12$, found using subtraction ($41-29=12$).

Teacher Checklist

- ☐ Explains addition and subtraction as inverse operations.
- ☐ Correctly finds a missing part using subtraction.
- ☐ Correctly finds a missing whole using addition.
- ☐ Completes a full fact family from 3 numbers.
- ☐ Uses inverse operations to check a worked answer.
- ☐ Applies the strategy to a worded problem.

Teacher Tip

Watch for students who always subtract regardless of what is missing. Reinforce the part-whole language: if the **whole** is missing, add the parts; if a **part** is missing, subtract the known part from the whole.